

# Zahra Golpayegani

AI Research Engineer  
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<https://zahragolpa.github.io/>

## About Me

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I am a machine learning research engineer with 5+ years of experience building **LLM post-training pipelines** and **scaling large language models** for **efficient training and inference**. My work spans RL post-training, hardware-aware optimization, distributed systems, and end-to-end ML infrastructure, with a focus on translating research into high-performance training and serving systems.

## Experience

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Huawei Canada – *Researcher*

Sep 2024 – Present

- Built an end-to-end **GRPO post-training pipeline** for Qwen-family models on Triton kernel generation, leveraging **VeRL** with **asynchronous rollouts** and **multi-turn tool use**.
- Built a **Uni-Agent-based RL environment** for Triton kernel generation, improving agent performance through **iterative optimization of skills, tool interfaces, prompts, and sandbox execution**.
- Integrated **asynchronous RL training with multi-turn tool use recipes** into **VeRL** for **Huawei Ascend NPUs**, extending **evaluation coverage** and improving **long-horizon** agent reasoning ([PR #101](#), [PR #102](#)).
- Implemented Google's **block-wise speculative decoding** in **vLLM** and **vLLM-Ascend**, improving **inference throughput by 10.2%** while **reducing TPOT by 9.2%** over standard speculative decoding ([PR #40819](#), [PR #7808](#)).
- Co-developed a simulator for **distributed LLM training and serving**, incorporating **kernel-level roofline models** to predict throughput and latency across hardware platforms and parallelization strategies.
- Implemented **Attention-FFN Disaggregation** (AFD) and **Prefill-Decode Disaggregation** (PDD) simulations for state-of-the-art LLM architectures (DeepSeek NSA, DSA, TurboS), enabling evaluation of **disaggregated serving pipelines**.
- Profiled **performance bottlenecks** in LLM training and serving workloads on **Huawei Ascend accelerators** through systematic **profiling**, and proposed targeted **parallelization strategy optimizations** (TP, EP, DP, etc.) to improve compute and memory efficiency for models such as **DeepSeek-NSA** and **MiniMax-Text-01** on Ascend NPUs.
- Accelerated LLaMA pre-training through conditional computation, **reducing training cost by 16%** while **increasing MFU by 3%** with no loss in model quality.

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Zetane Systems – *AI Engineer*

Dec 2021 – Aug 2024

- Co-designed and developed [ZetaForge](#), an AI platform automating prompt-to-pipeline generation using LLMs.
- Developed complex end-to-end multi-agent AI pipelines leveraging state-of-the-art libraries such as **LangChain** and **openai-python** ([blog post](#)).
- Implemented a robust **MLOps pipeline** (end-to-end machine learning operations workflow) featuring data and model validation, preprocessing, model training with **automated experiment tracking**, and continuous model improvement ([open-source project](#)).

- Contributed to flagship products including Protector—a platform for assessing vision model reliability in real-world conditions.
- Delivered product demos and technical talks to prospective clients and international audiences at events including Scale AI, World Summit AI, and ALL IN.

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**XAI Lab, Concordia University** – *Research Assistant*

Sep 2021 – Jan 2024

*Tools: Python, PyTorch, OpenCV, MATLAB, LaTeX*

- Investigated the interplay between **model robustness, accuracy, and shape bias** under Professor Nizar Bouguila’s supervision; published findings at **CRV 2023**.
- Proposed a novel image compression algorithm inspired by **Singular Value Decomposition (SVD)** to optimize robustness of vision models trained on compressed data; work published at **ICPRAM 2024**.

## Skills

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**Programming Languages:** Python, C/C++, CUDA

**Deep Learning Frameworks:** PyTorch, vLLM, VeRL, Megatron-LM, HuggingFace, LLaMA Factory

**MLOps & DevOps:** Docker, LangChain, MLflow, Git

## Education

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**Concordia University**, MSc Information Systems Engineering

2024

GPA: 4.0 / 4.0

Thesis: Enhancing Deep Learning Model Robustness: Insights from Out of Distribution Data Augmentation and an Innovative Image Compression Technique

Supervisor: Prof. Nizar Bouguila

**Amirkabir University of Technology**, BSc Computer Science

2020

GPA: 3.7 / 4.0

Final Project: Food9K: Detecting Food in Social Media Images Using YOLOv3

Supervisor: Prof. Mohammad Akbari

## Publications

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- ETT: Expanding the Long Context Understanding Capability of LLMs at Test-Time. 2025. arXiv. ([link](#))
- PatchSVD: A Non-Uniform SVD-Based Image Compression Algorithm. 2024. ICPRAM. ([link](#))
- Clarifying myths about the relationship between shape bias, accuracy, and robustness. 2023. CRV. ([link](#))

## Awards

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- **Campaign Hero Award**, Huawei 2025  
Honored for delivering rapid, high-impact results on a key internal campaign at Huawei HQ.
- **Future Star Award**, Huawei 2025  
Recognized for outstanding performance and potential, highlighting my contributions.
- **Best Paper Award Candidate**, ICPRAM 2024  
Nominated for research on model compression and robustness.
- **Conference and Exposition Award**, Concordia University 2023  
Awarded for excellence in presenting graduate-level research.
- **Accelerate Explore Award**, Mitacs 2021  
For conducting applied AI research in collaboration with industry partners.
- **Merit Scholarship Entrance Award**, Concordia University 2021  
Awarded for high academic achievement upon admission to MASc program.
- **2<sup>nd</sup> Place, Worldwide**, RoboCup Junior Soccer League 2013  
RoboCup World Competitions – international robotics tournament.